

7. (a) Solve $\begin{vmatrix} 5-x & -3 \\ 4 & -2-x \end{vmatrix} = 0$ (*). (2 marks)

(b) Let α_1 and α_2 ($\alpha_1 < \alpha_2$) be the solutions of (*), and $P = \begin{pmatrix} a & b \\ 1 & -1 \end{pmatrix}$.

It is given that $\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} a \\ 1 \end{pmatrix} = \alpha_1 \begin{pmatrix} a \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} b \\ -1 \end{pmatrix} = \alpha_2 \begin{pmatrix} b \\ -1 \end{pmatrix}$,

where a and b are constants.

(i) Find P .

(ii) Evaluate $P^{-1} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} P$.

(iii) Using (b)(ii), evaluate $\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix}^{23}$. (6 marks)

a) $(5-x)(-2-x) - (-3)(4) = 0$ IM
 $-10 + 2x - 5x + x^2 + 12 = 0$
 $x^2 - 3x + 2 = 0$
 $x = 2$ or $x = 1$ IA

b i) $\alpha_1 = 1, \alpha_2 = 2$

$\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} a \\ 1 \end{pmatrix} = \alpha_1 \begin{pmatrix} a \\ 1 \end{pmatrix}$	$\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} b \\ -1 \end{pmatrix} = \alpha_2 \begin{pmatrix} b \\ -1 \end{pmatrix}$
$\begin{pmatrix} 5a-3 \\ 4a-2 \end{pmatrix} = \begin{pmatrix} a \\ 1 \end{pmatrix}^*$	$\begin{pmatrix} 5b+3 \\ 4b+2 \end{pmatrix} = \begin{pmatrix} 2b \\ -2 \end{pmatrix}^*$ IM for either
$a = \frac{3}{4}$	$b = -1$

$\therefore P = \begin{pmatrix} \frac{3}{4} & -1 \\ 1 & -1 \end{pmatrix}$ IA

ii) $P^{-1} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} \frac{3}{4} & -1 \\ 1 & -1 \end{pmatrix}$

$= \frac{1}{(-1)(\frac{3}{4}) - (-1)(1)} \begin{pmatrix} -1 & 1 \\ -1 & \frac{3}{4} \end{pmatrix} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} \frac{3}{4} & -1 \\ 1 & -1 \end{pmatrix}$ IM for P^{-1}

$= 4 \begin{pmatrix} -1 & 1 \\ -2 & \frac{3}{2} \end{pmatrix} \begin{pmatrix} \frac{3}{4} & -1 \\ 1 & -1 \end{pmatrix}$

$$= 4 \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{IA}$$

$$\text{ii) } \left(P^{-1} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix} P \right)^{23} = P^{-1} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix}^{23} P$$

$$P^{-1} \begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix}^{23} P = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^{23}$$

$$\begin{pmatrix} 5 & -3 \\ 4 & -2 \end{pmatrix}^{23} = \begin{pmatrix} \frac{3}{4} & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1^{23} & 0 \\ 0 & 2^{23} \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ -1 & \frac{3}{4} \end{pmatrix} \quad \text{IM}$$

$$= 4 \begin{pmatrix} \frac{3}{4} & -2^{23} \\ 1 & -2^{23} \end{pmatrix} \begin{pmatrix} -1 & 1 \\ -1 & \frac{3}{4} \end{pmatrix}$$

$$= 4 \begin{pmatrix} -\frac{3}{4} + 2^{23} & \frac{3}{4} - \frac{3}{4}(2^{23}) \\ -1 + 2^{23} & 1 - \frac{3}{4}(2^{23}) \end{pmatrix}$$

$$= \begin{pmatrix} -3 + 2^{25} & 3 - 3(2^{23}) \\ -4 + 2^{25} & 4 - 3(2^{23}) \end{pmatrix} \quad \text{IA}$$